

Spoken Tutorial – Code and Audio Generation: AI's Specialized Skills

Title: Code and Audio Generation: AI's Specialized Skills

Module	Generative AI Beyond Text and Images
Episode	4.3
Learning Objective	At the end of this tutorial learner will be able to 1. Understand how generative AI assists in code creation. 2. Identify key applications of AI in audio generation. 3. Recognize the benefits of AI in specialized domains like coding and sound.
Approx. Duration	Approximately 10-12 minutes
Outline	• Welcome to Code and Audio Generation. • Learning Objectives for this tutorial. • System Requirements to follow along. • Prerequisites: What you should know beforehand. • AI's Expanding Creativity: Beyond Text and Images. • Introducing Code Generation with AI. • How AI Tools Enhance Developer Productivity. • Code Generation: Your Smart Developer Assistant. • Exploring Audio Generation with AI. • Voice Synthesis and Text-to-Speech Applications. • AI for Music Creation and Composition. • Audio Generation: The AI Sound Wizards. • Summary of what we learned. • Assignment: Practice your understanding. • Thank you and Acknowledgements.
Meta Tags	generative AI, AI in coding, GitHub Copilot, code generation, AI music, audio generation, voice synthesis, text-to-speech
Pre-requisite Tutorial	Familiarity with basic concepts of Artificial Intelligence and Machine Learning. Understanding of how AI works with text and images.

Script

Visual Cue	Narration
A futuristic banner with 'Code and Audio Generation' prominently displayed, showing abstract representations of code lines and sound waves.	Welcome to this Spoken Tutorial. It is about Code and Audio Generation.

A bulleted list icon with gears and sound waves, representing code and audio, next to a checklist.	In this tutorial, you will learn many things. You will understand AI in code creation. You will identify AI 's audio applications. You will recognize AI 's benefits in these domains.
Icons representing a web browser, Wi-Fi signal, and a desktop computer.	You will need a web browser. Use Firefox or Chrome. You need a stable internet connection. No software installation is needed.
A stack of books or building blocks representing foundational knowledge, with 'AI Basics' written on them.	To follow this tutorial, you must know AI basics. You should know Machine Learning concepts. You must understand AI with text and images.
A robot hand typing code on one side and a robot hand conducting an orchestra on the other, symbolizing AI's reach into new domains.	Have you spent hours debugging code? Think about it. Do you need music for a video? Is it hard to find the right track? Generative AI is mastering complex domains. It goes beyond text and images. It steps into programming and sound.
A developer working at a computer, with a holographic AI assistant figure peering over their shoulder, suggesting code.	Imagine a coding partner. It knows billions of code lines. Tools like GitHub Copilot exist. They are AI models. They train on vast public code. They act as a smart assistant. They do not replace human programmers.
A split screen showing a code editor. On one side, a human types, and on the other, AI suggestions appear, with arrows indicating increased speed and efficiency.	How does this AI assistant help you? Think about it. It suggests code as you type. It translates comments into code. It helps you find bugs faster. It helps you fix bugs faster. This makes developers quicker. It makes them more efficient.
Two people (one human, one stylized AI figure) sitting side-by-side at a computer, collaboratively coding.	Think of AI code generation. It is like a pair programmer. This programmer is highly experienced. It knows common patterns. It knows common solutions. It anticipates your needs. It offers suggestions quickly. It is an encyclopedia of code. It guides you constantly.

Abstract sound waves transforming into recognizable audio elements like a microphone, musical notes, and headphones.	AI masters more than visuals. It masters more than text. AI is now a master of sound. These models train on vast datasets. They use sounds and human voices. They use diverse music. They learn audio patterns. They generate audio patterns.
A microphone icon with sound waves emanating from it, transforming into text bubbles for audiobooks and a virtual assistant avatar.	What can AI do with audio? Think about it. It creates realistic voices. These voices are human-like. This is voice synthesis. It is also text-to-speech. Use it for audiobooks. Use it for podcasts. Use it for virtual assistants. The voices sound real.
A musical staff with AI elements, notes flowing from a computer screen, and a diverse range of musical instruments.	AI can also compose music. It generates background music. This music is royalty-free. Use it for videos. Use it for presentations. It creates new melodies. It creates new harmonies. Some AI models generate songs. They match a famous artist's style. They match a specific genre.
A stylized wizard figure with musical notes and sound waves swirling around its hands, creating a magical auditory scene.	Consider AI audio generation. It is a team of sound wizards. They conjure any voice you need. The voices are perfectly articulate. They compose entire symphonies. They do it on demand. It is a virtual recording studio. It is a master composer. Both are available instantly.

A concise infographic summarizing code generation and audio generation with icons for each.	We explored generative AI. It extends to code and audio. Code generation tools are smart assistants. GitHub Copilot is one example. They boost developer efficiency. They suggest code. They find bugs. Audio generation creates realistic voices. This is voice synthesis. It composes new music. It creates soundscapes.
A developer at a computer screen, with a thought bubble showing lines of code being written automatically by an AI hand.	Think about a coding task. Do you perform it often? Pause and consider. Research a free AI code tool. Find a plugin for your IDE. Try using it for that task. Automate it or get assistance. Reflect on your workflow. Did it change your efficiency?
The EduPyramids Educational Services Private Limited and SINE, IIT Bombay logos.	This Spoken Tutorial comes from EduPyramids. It is also from SINE, IIT Bombay. Thank you for joining us.